SYSTEM DESIGN

CRDTs
Visual Guide

Conflict-free Replicated Data Types:
convergence without coordination.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Convergence Without Coordination

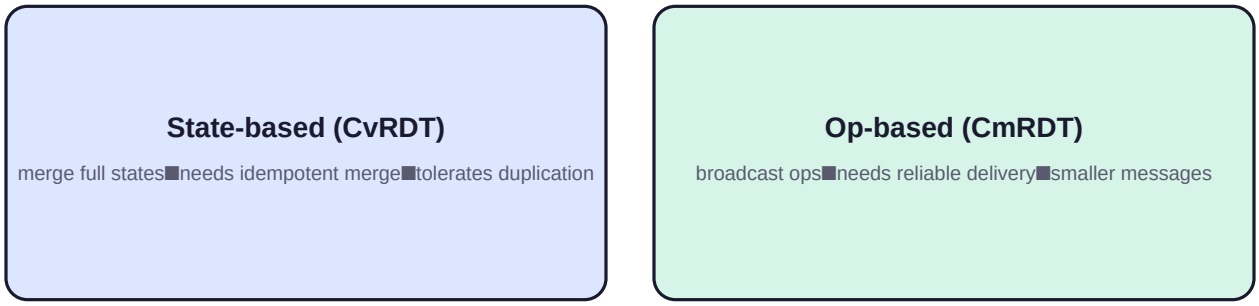
Replicas merge whenever they meet -- and ALWAYS agree, with no leader, no locks.

- The math requirement: merge must be commutative, associative, idempotent -- order and duplicates do not matter.
- Offline-first dream: edit on a plane, sync later, no conflicts EVER (for CRDT-shaped data).
- Price: only certain data shapes qualify -- counters, sets, registers, text. Not arbitrary SQL.

STEP 2/6

State-Based vs Operation-Based

Ship the whole state, or ship just the operation?



- State-based: simple and robust (Riak-style) -- but state can be large to ship.
- Op-based: tiny messages ('increment by 1') -- but the channel must deliver each op exactly once, in causal order.
- Delta-state: the modern hybrid -- ship only the CHANGED part of the state. Best of both.

STEP 3/6

G-Counter: The Simplest CRDT

Grow-only counter: one slot per replica, value = sum of slots.

```
A: [2, 0, 0]      B: [0, 1, 0]      merge -> [2, 1, 0]  =  3
```

- Each replica increments ONLY its own slot -- no coordination, no lost updates.
- Merge = take max per slot: concurrent increments never overwrite each other.
- PN-Counter adds a second grow-only counter for decrements: value = plus - minus.
- Use: like counts, vote tallies, retry-safe metrics -- anywhere 'count up' is the job.

STEP 4/6

Sets: Harder Than They Look

Add is easy. Remove + concurrent add = the classic CRDT puzzle.

G-Set

add-only■merge = union■trivially safe

2P-Set

add set + tombstone set■remove wins forever■cannot re-add!

- 2P-Set flaw: once removed, an element can NEVER return -- the tombstone outlives everything.
- OR-Set (observed-remove): tag each ADD uniquely; remove deletes only observed tags -- re-add works.
- Rule: 'remove' must mean 'remove what I have seen', never 'remove forever'.

STEP 5/6

LWW Registers

Last-writer-wins: attach a timestamp, highest wins -- simple, lossy, honest about it.

```
A sets name='Al' @t=5      B sets name='Bo' @t=7

merge -> 'Bo' (higher timestamp wins, 'Al' is LOST)
```

- Timestamps must be trustworthy: hybrid logical clocks beat wall clocks here.
- Data loss is BY DESIGN: concurrent writes to one register cannot both survive.
- Use for: display names, status fields -- single values where 'latest intention' is the semantics.
- Need both values kept? That is a multi-value register -- see vector clocks + siblings.

STEP 6/6

Collaborative Editing

Google Docs / Figma: CRDTs (or OT) under the hood.

Characters are list elements with stable IDs; concurrent inserts interleave deterministically.

- Both editors converge to the SAME document with no server merging logic -- the data structure does it.
- Yjs and Automerge package this: local-first apps with sync as a side effect.
- OT (operational transform) is the older cousin: needs a central server to transform ops; CRDTs do not.

STEP ✓I✓

Key Takeaways

CRDTs — Remember

- CRDTs guarantee replicas converge with no coordination: merge must be commutative, associative, idempotent.
- State-based ships full state (robust); op-based ships operations (small); delta-state splits the difference.
- G-Counter: per-replica slots, merge by max -- concurrent increments never lost.
- Sets need care: OR-Set's tagged adds make remove-then-re-add work; 2P-Set forbids re-adding.
- LWW registers keep the latest timestamp (lossy by design); sequence CRDTs power collaborative editors.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev